

Parallel operation of the CS1070 power amplifier

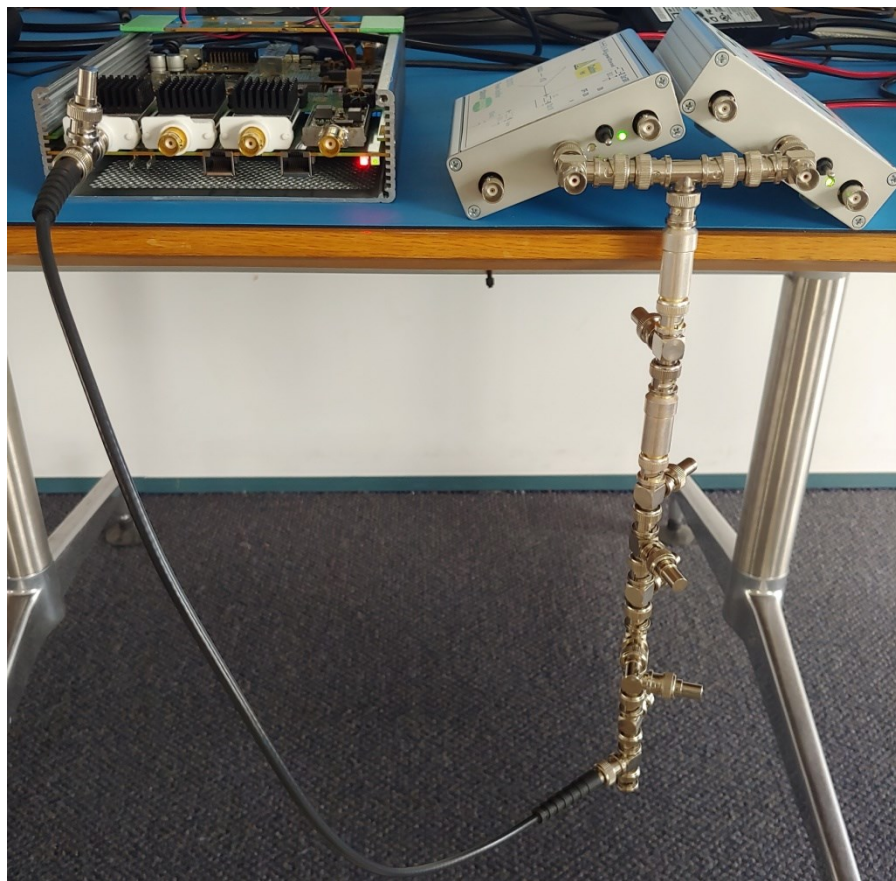
Summary

The CS1070 1 ohm output includes a series 1 ohm resistor. This allows two CS1070's to be paralleled, and current to be shared between them. This brings advantages in total output current, and reduced distortion at higher output levels.

It is **vital**ly important that the two signals driving the CS1070's are exactly the same - with the same amplitude and phase. Any imbalance will lead to current flow between the two units, decreasing efficiency, and increasing distortion.

Setup

A development CS448 was used for measurements (which is why it is open). The sig gen filtered output went via a T splitter through two equal length BNC cables to the two CS1070's. The load is therefore 25 ohms, and the phase and amplitude presented to the CS1070's close to identical.



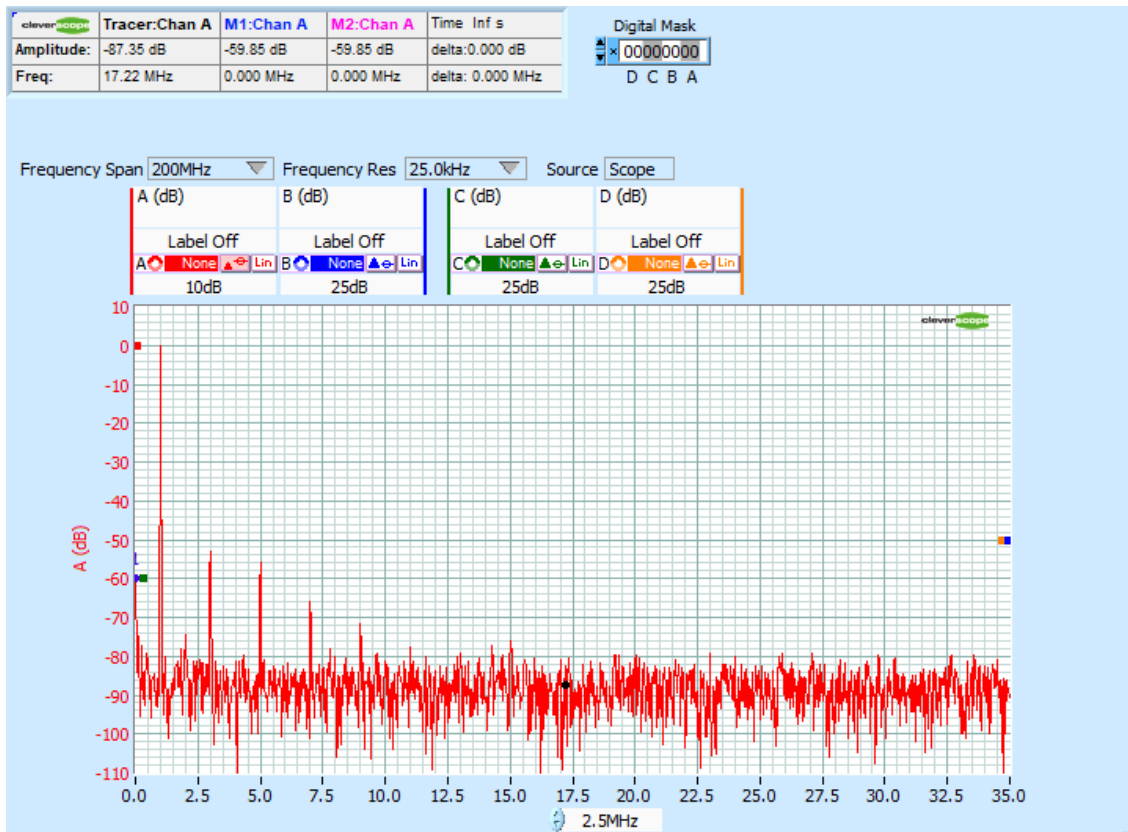
The two CS1070's outputs were connected with coaxial T connectors, and a tree of 50 ohm loads - 10 in all. The load impedance is thus 5 ohm. A couple are through loads.

The CS1070's were placed for best heat dissipation.

Measurements

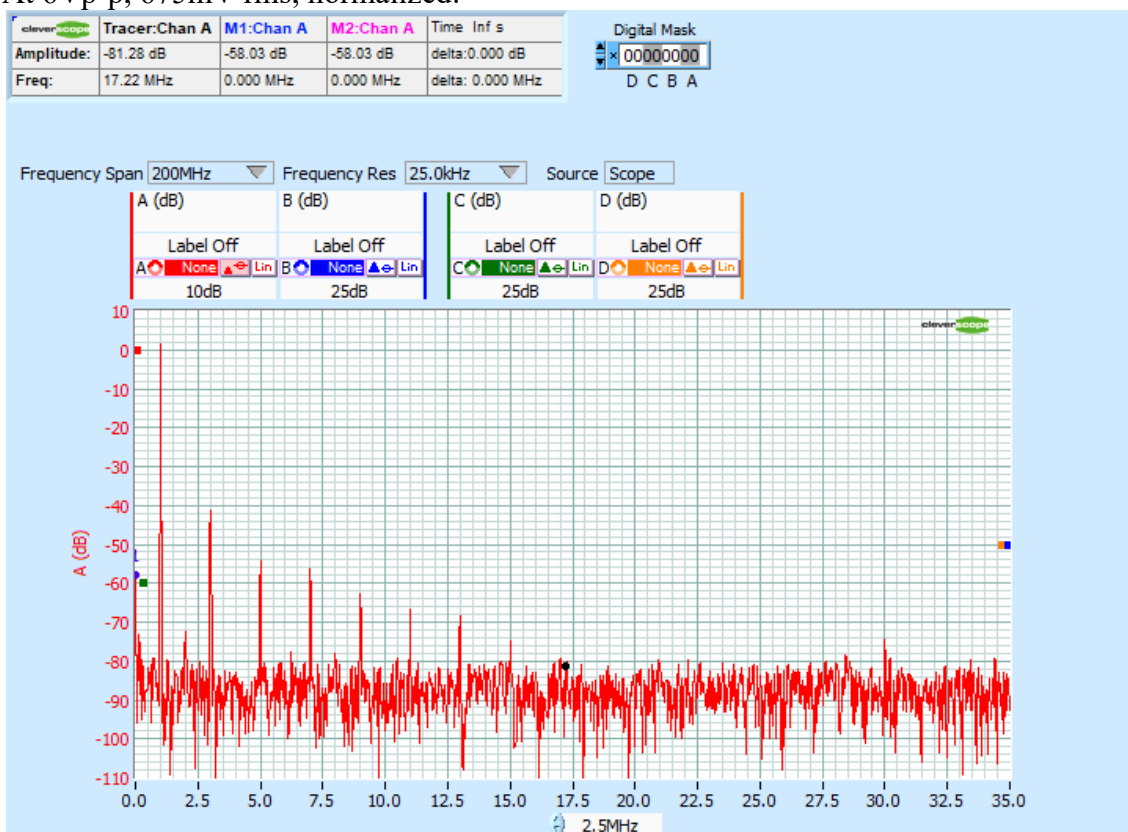
Characterizing the Signal Generator

Here we measured the Signal Generator output alone, into 25 ohm (same as two paralleled CS1070's). The Sig Gen filtered output was used. Using a sig gen voltage of 5V p-p open but loaded the with 25 ohm, measuring 565mV rms at the scope input, this is the distortion seen by the input of the CS1070 at 1MHz, with the fundamental normalized:



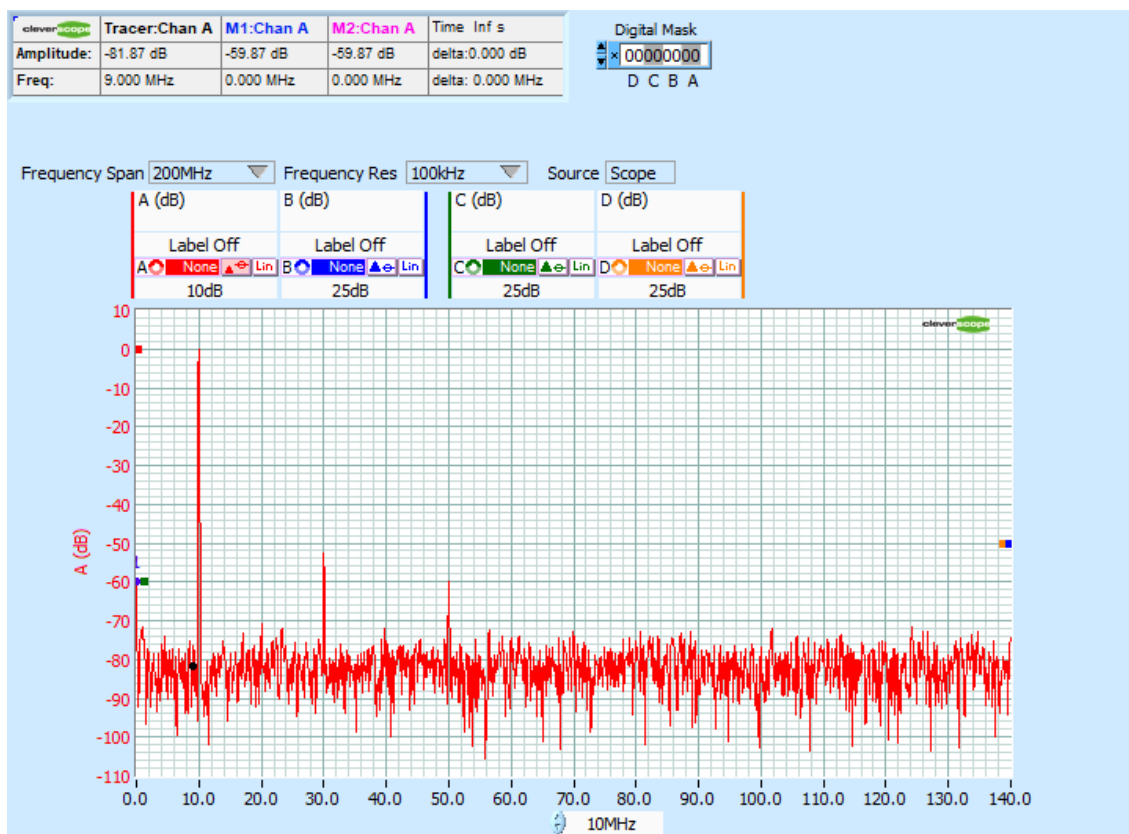
Distortion is about -55 dBc, and the 3rd harmonic is lower than the 2nd harmonic,

At 6Vp-p, 675mV rms, normalized:



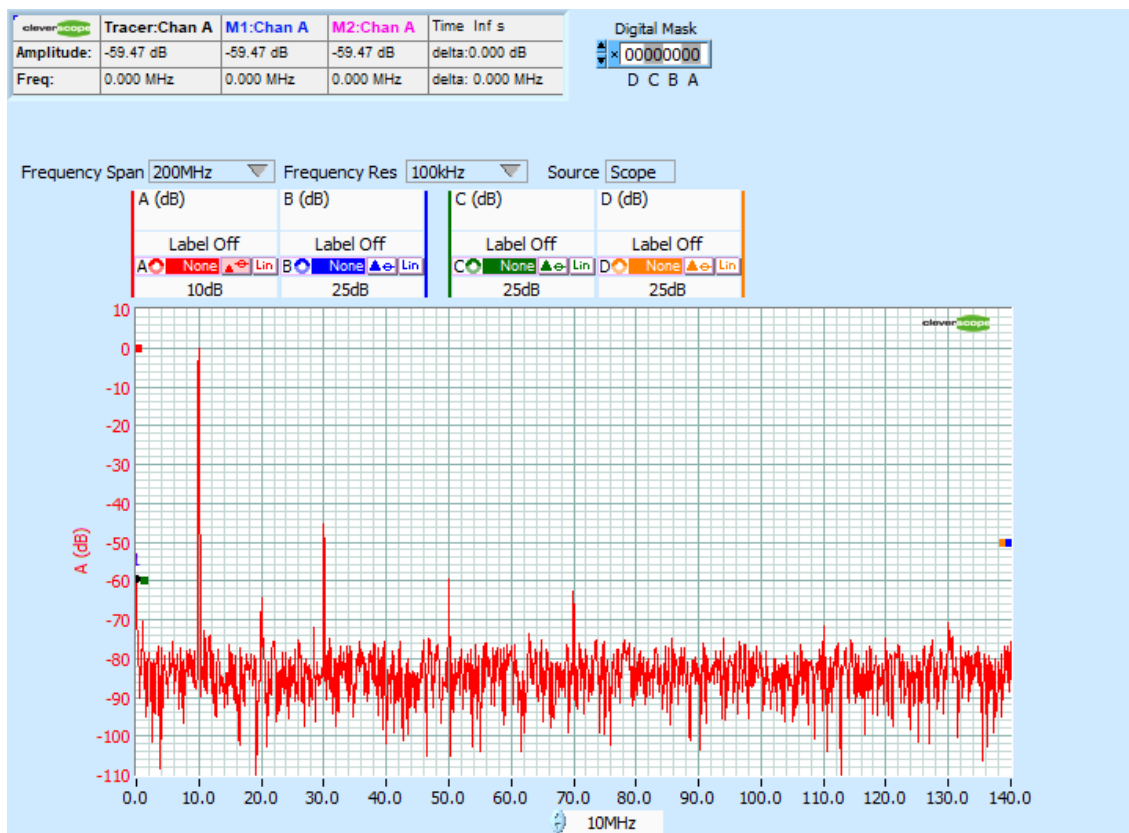
So distortion is up, and now -41 dBc, for the 2nd harmonic.

At 10 Mhz, 5 V, 552mV rms measured:



Distortion is about -52dBc for the 2nd harmonic.

And 6V, 660mV rms measured :

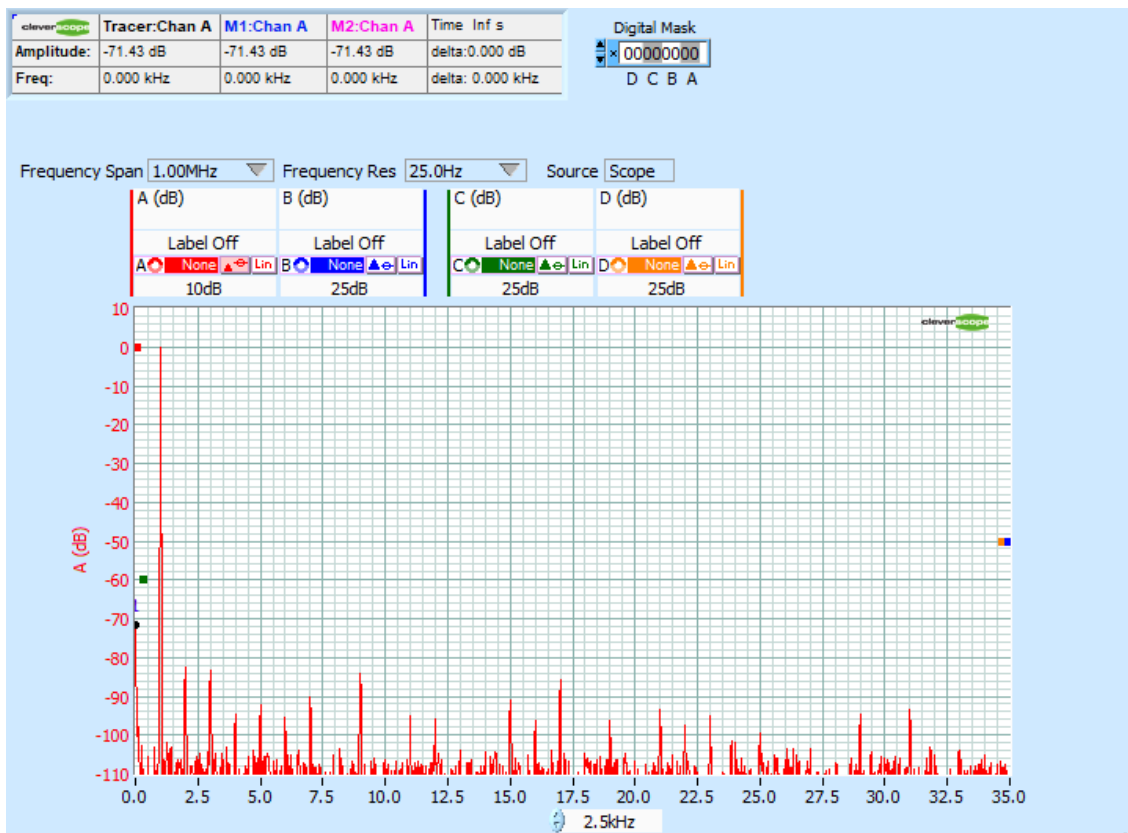


Distortion has increase to -45dBc for the 2nd harmonic.

Measurements of paralleled CS1070's.

Using equal length Coaxial cables from both CS1070's to a T connector on CS448 output.
Equal length T arrangement on outputs, with 5 ohm load tree.

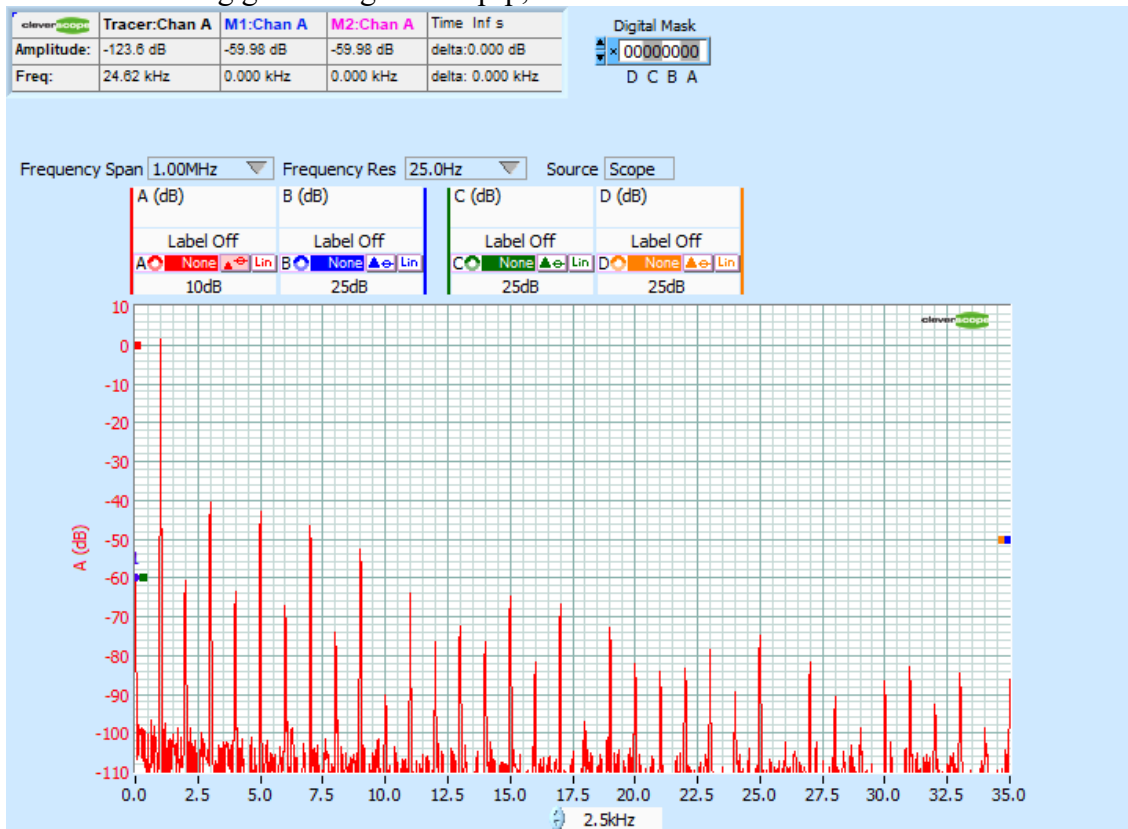
At 1kHz, with 5V p-p indicated out of the sig gen (into 25 ohm, it will be $\frac{1}{4}$ of this actual), 5.1V rms measured on the CS1070 1 ohm output (with 10x gain), the Spectrum response (normalized) is:



Current sharing was good, with CS1070-1 using 557 mA and CS1070-2 using 560mA.

Distortion is below -80 dBc.

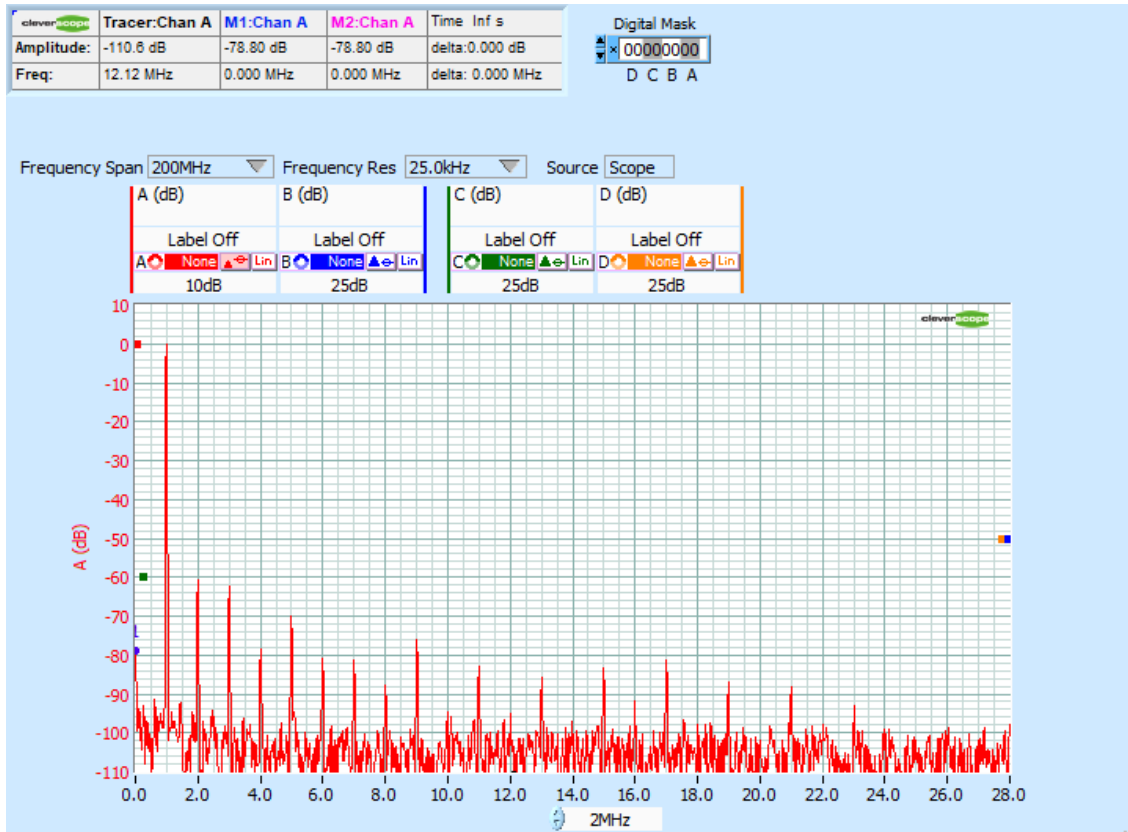
I increased the sig gen voltage to 6V p-p, 6.07V rms.



Current sharing was still good at about 656/659 mA each.

Distortion was now higher at -40 dB, this is due to the sig gen output distortion, combined with the CS1070 unbalance distortion.

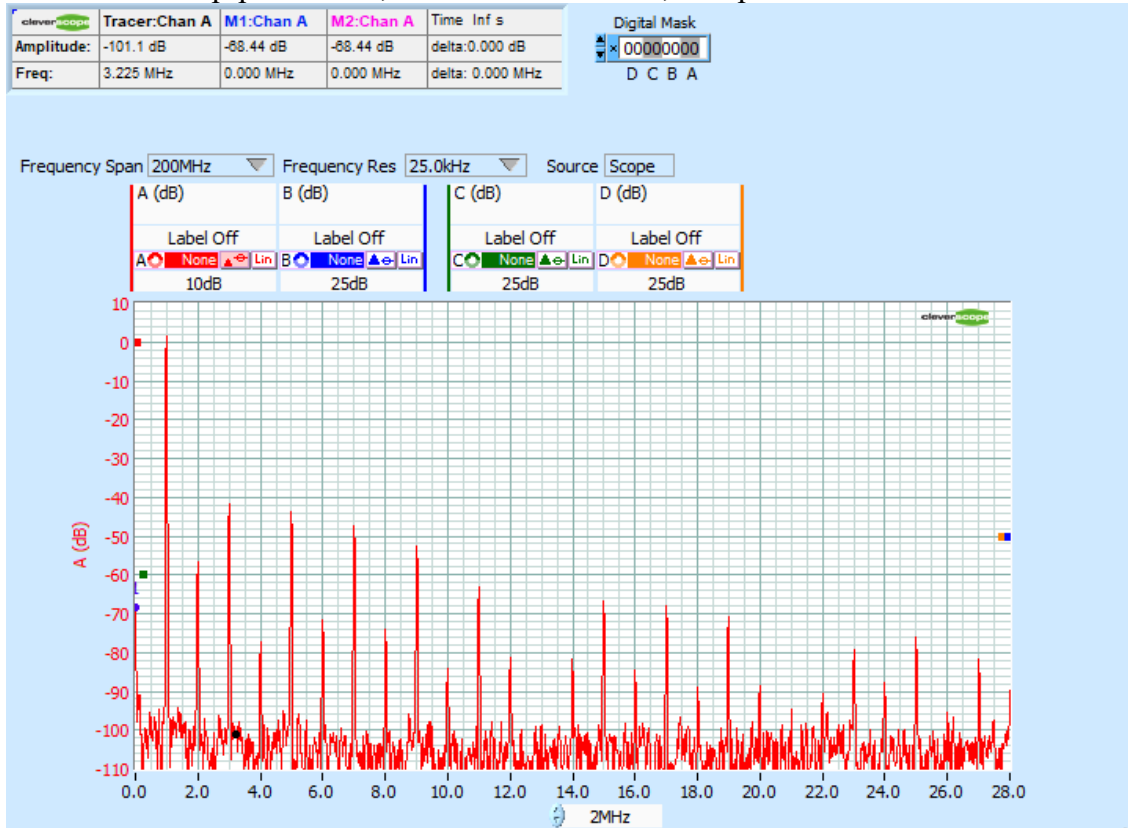
With 1MHz, 5V p-p indicated, 5.1V RMS measured, the spectrum is:



Current sharing is 560/5625 ma per unit.

Distortion is better than -60 dBc

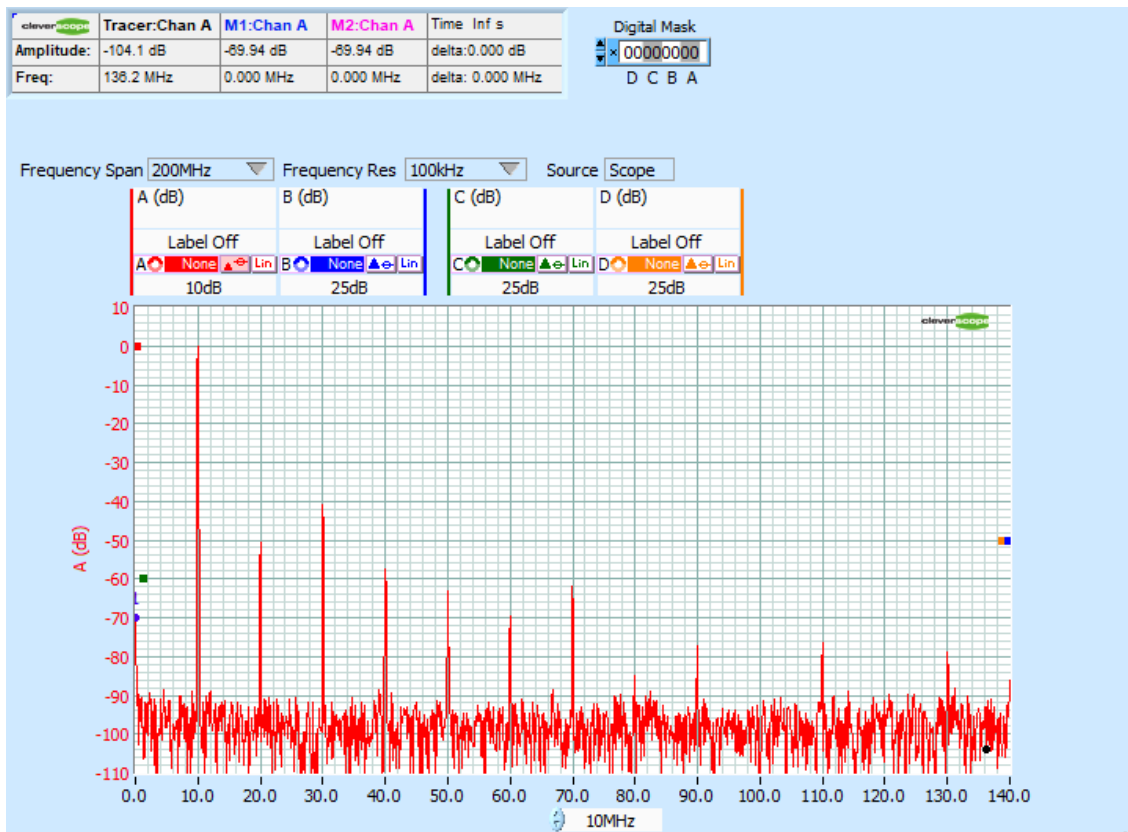
Increased to 6V p-p indicated, 6.07V rms measured, the spectrum is:



Current sharing is good at 656/659 mA per unit.

We see the same -40dBc as the sig gen by itself.

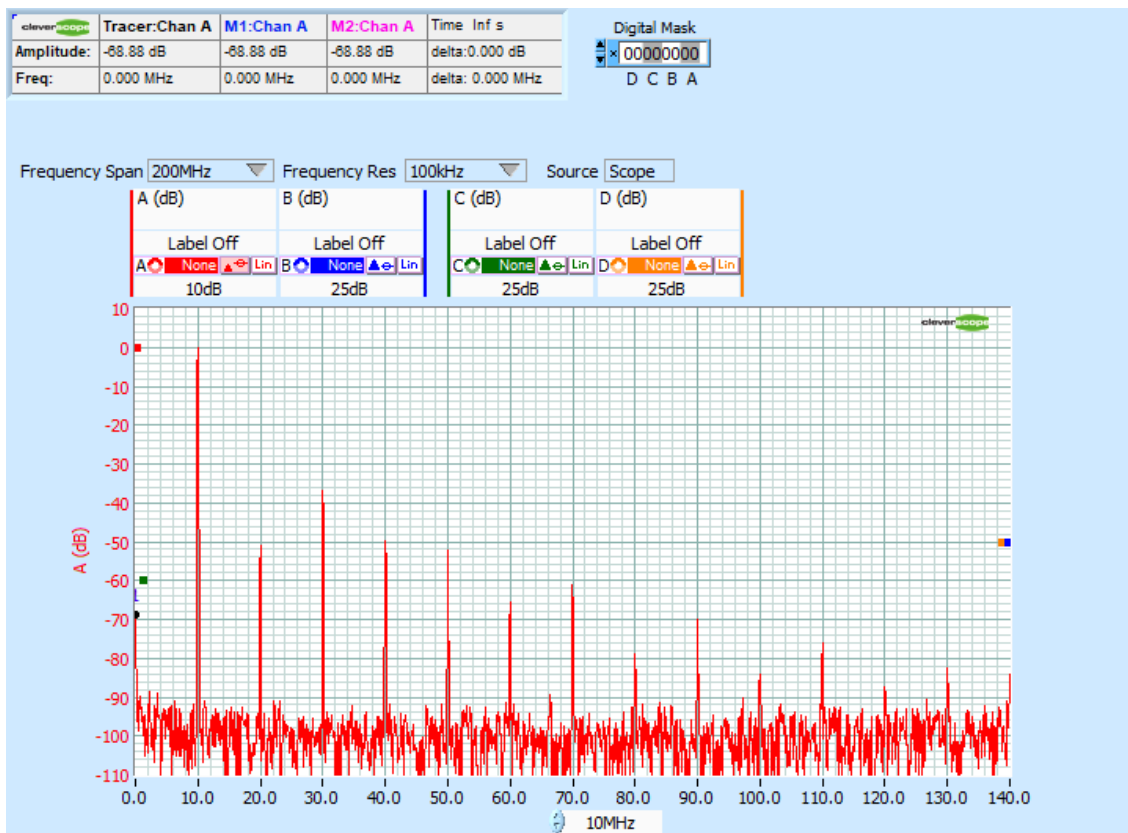
With 10 Mhz, 5V p-p indicated, 4.05V rms measured, the spectrum is:



The current was 493 and 453mA respectively. This is the first imbalance I see.

Distortion increased with 3rd order distortion of -40dBc, worse than the sig gen on its own.

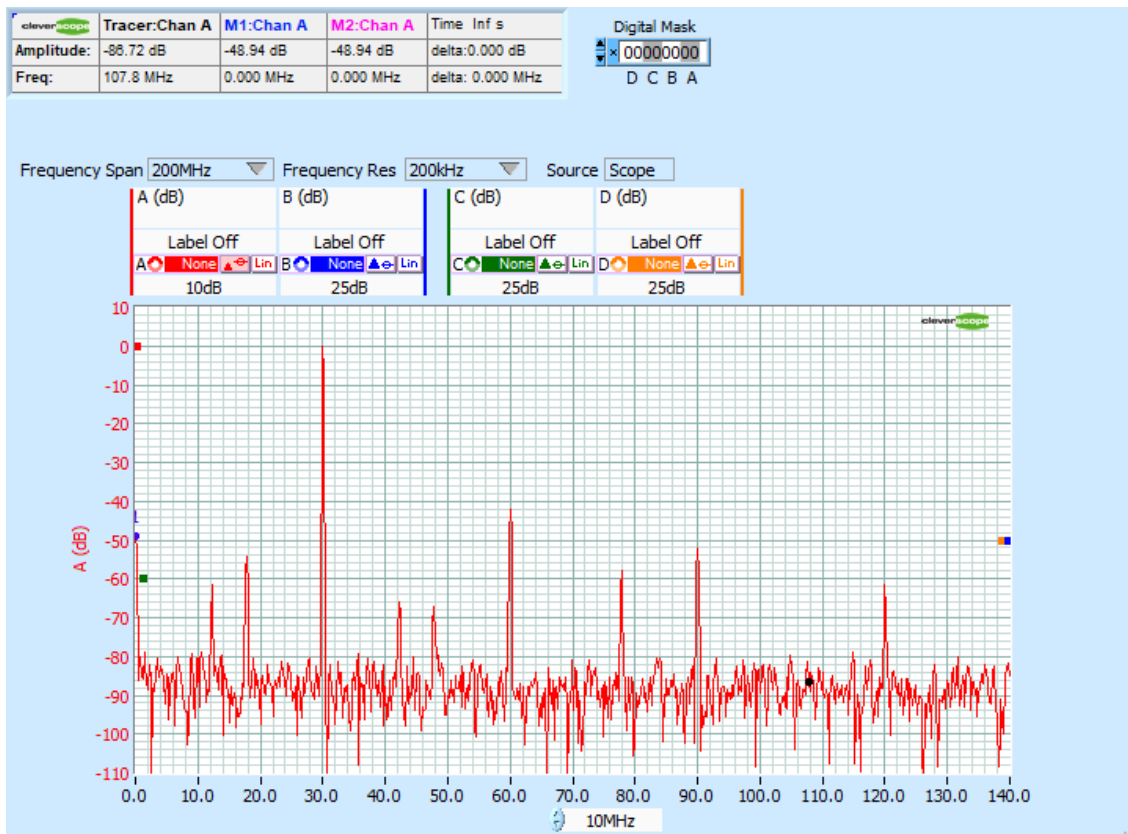
At 6V indicated, 4.86V rms, the spectrum is:



The current used is unbalanced with 584mA and 532 mA respectively.

There is increased 3rd order distortion to -36dBc.

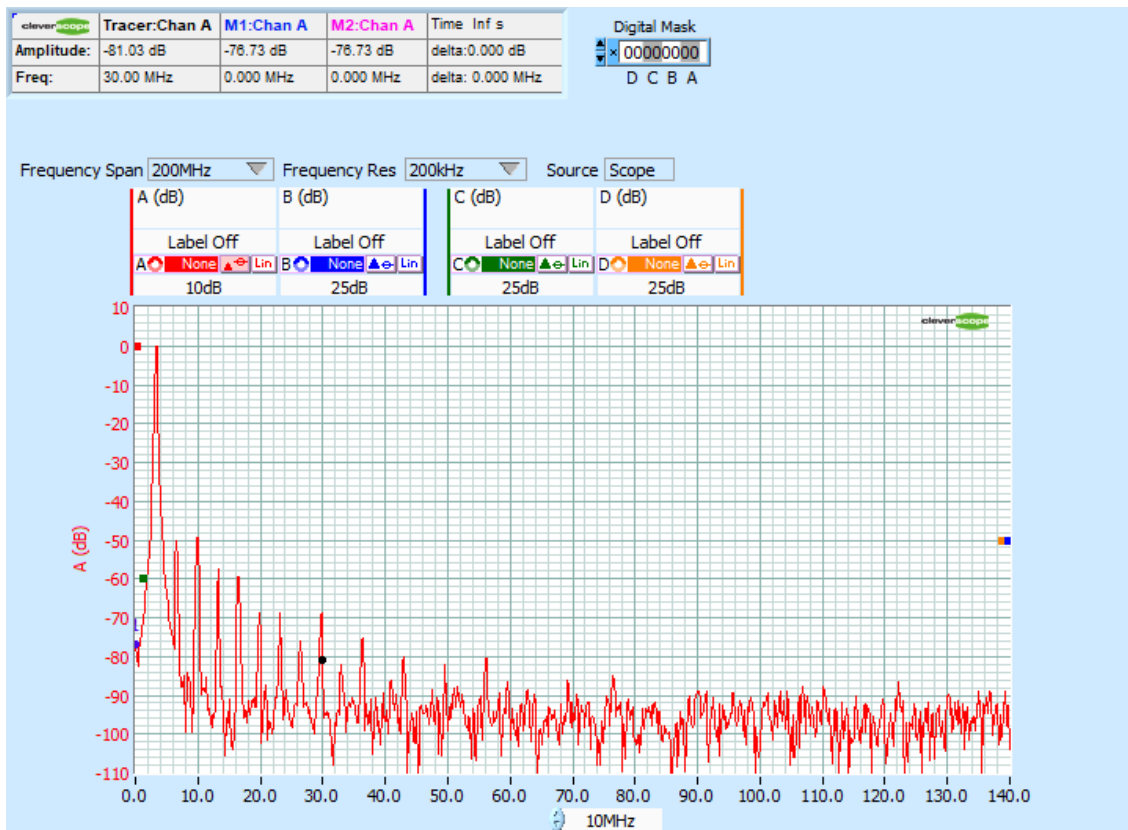
At 30 Mhz, we get a reduction in amplitude due to the sig gen filter, and power amplifier bandwidth. With 5V p-p, measured 1.754V rms, the normalized spectrum is:



The CS1070 input currents are 281 and 251 mA respectively.

Additional peaks are intermod products between DDS sample frequency and harmonics of the 30 MHz and unbalanced products.

I explored the frequency at which the 3rd harmonic is first larger than the 2nd (indicating current imbalance) and it is 3.3MHz. With 5V sig gen V, 4.91V rms measured, the spectrum is:

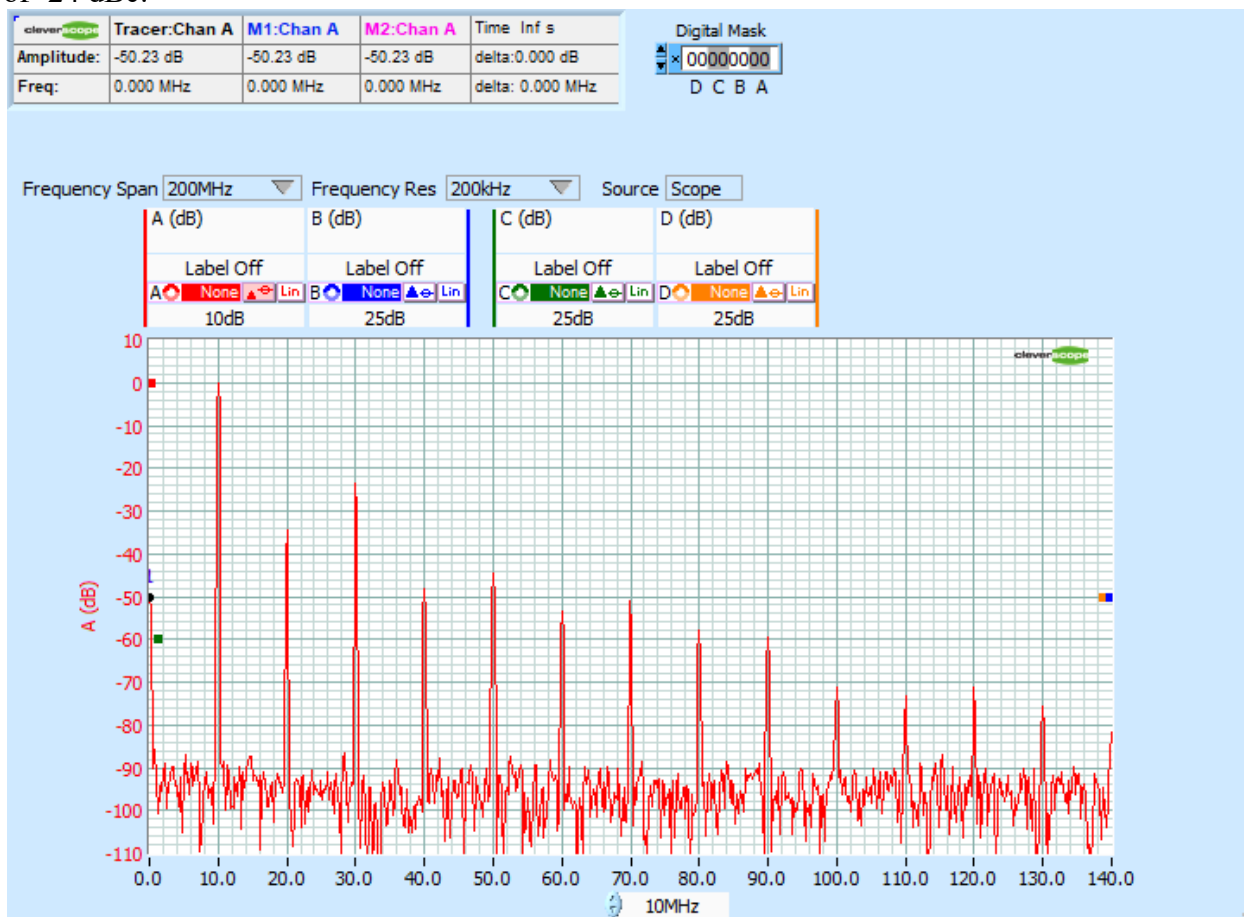


CS1070 currents are 555/542mA.

Above this frequency, current imbalance grows, but is still useful up to at least 10 MHz (1% distortion).

This crossover is probably due to small imbalances in the gain of the CS1070's.

With a single CS1070, and the same 5 ohm load, current was 0.549A at 10 MHz, and -40 dBc distortion, amplitude was 2.417V rms. At 4.05Vrms, 1A input current, distortion was much worse with a 3rd harmonic of -24 dBc:



Conclusion

Provided the amplitude and phase of the signals input to the two CS1070's are exactly the same, the CS1070's will current share well with low distortion up to about 10 MHz. This figure may be improved by normalizing the output amplitude and phase for each CS1070 to be the same. Cleverscope may provide a calibration panel option to do this using the dual output signal generator in the future, if there is demand.